

Wine Dataset – QC Report

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WINE DATASET QUALITY CONTROL REPORT
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.. 1. STRUCTURE .....
Rows: 20 | Columns: 12
Column names & types:
  name          character
  country        character
  region         character
  grape          character
  color          character
  vintage        integer
  alcohol        numeric
  acidity         numeric
  residual_sugar numeric
  ph             numeric
  price_eur      numeric
  rating         integer

.. 2. MISSING VALUES .....
PASS: No missing or blank values found.

.. 3. DUPLICATE ROWS .....
PASS: No duplicate rows.
PASS: All wine names are unique.

.. 4. CATEGORICAL VALUES .....
PASS: All color values valid.
Color distribution: red=10  rose=1  sparkling=2  white=7
Countries (12): Argentina, Australia, Austria, France, Germany, Hungary, Italy, New Zealand, Portugal, South Africa, Spain, USA
Grapes (17): Cabernet Sauvignon, Chardonnay, Chenin Blanc, Furmint, Gewurztraminer, Grenache, Gruner Veltliner, Macabeo, Malbec, Merlot, Pinot Noir, Riesling, Sangiovese, Sauvignon Blanc, Shiraz, Temp

.. 5. NUMERIC RANGE CHECKS .....
PASS  alcohol      [5.0 - 22.0]
PASS  acidity       [3.0 - 12.0]
PASS  residual_sugar [0.0 - 400.0]
PASS  ph            [2.5 - 4.5]
PASS  price_eur     [0.0 - 10000.0]
PASS  rating        [50.0 - 100.0]
PASS  vintage       [1900.0 - 2026.0]
All numeric ranges OK.

.. 6. OUTLIER DETECTION (IQR × 1.5) .....
WARN  alcohol      fence [9.90, 16.50]
      Row 3  Mosel Kabinett      = 8.5  (LOW)
WARN  acidity       fence [3.65, 9.05]
      Row 15 Tokaji Aszu         = 9.5  (HIGH)
WARN  residual_sugar fence [-3.38, 10.62]
      Row 3  Mosel Kabinett      = 38   (HIGH)
      Row 15 Tokaji Aszu         = 180  (HIGH)
      Row 19 Alsace Gewurz       = 12   (HIGH)
WARN  price_eur     fence [-11.62, 69.38]
      Row 9  Napa Reserve        = 85   (HIGH)

.. 7. DOMAIN CONSISTENCY CHECKS .....
INFO  1 dessert-style wine(s) (residual sugar >45 g/L):
      Tokaji Aszu          sugar=180 g/L  color=white
PASS  Red wine alcohol levels all >=11%.
PASS  No high-price / low-rating inconsistencies.

.. 8. SUMMARY STATISTICS .....
alcohol      min= 8.50  Q1= 12.38  median= 13.50  mean= 13.03  Q3= 14.03  max= 15.00  sd= 1.53
acidity       min= 5.20  Q1= 5.68   median= 6.15   mean= 6.52   Q3= 7.03   max= 9.50   sd= 1.18
residual_sugar min= 1.20  Q1= 1.87   median= 2.25   mean= 14.01  Q3= 5.38   max= 180.00 sd= 39.93
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ph	min=	3.05	Q1=	3.27	median=	3.45	mean=	3.42	Q3=	3.58	max=	3.70	sd=	0.20
price_eur	min=	13.00	Q1=	18.75	median=	25.25	mean=	31.70	Q3=	39.00	max=	85.00	sd=	18.52
rating	min=	85.00	Q1=	88.00	median=	89.50	mean=	89.85	Q3=	92.00	max=	95.00	sd=	2.74

.. 9. CORRELATION MATRIX

	alcohol	acidity	residual_sugar	ph	price_eur	rating
alcohol	1.00	-0.88	-0.46	0.83	0.20	0.10
acidity	-0.88	1.00	0.69	-0.74	0.06	0.17
residual_sugar	-0.46	0.69	1.00	-0.13	0.26	0.43
ph	0.83	-0.74	-0.13	1.00	0.23	0.26
price_eur	0.20	0.06	0.26	0.23	1.00	0.85
rating	0.10	0.17	0.43	0.26	0.85	1.00

Notable correlations ($|r| > 0.6$):

alcohol	~ acidity	r = -0.88
alcohol	~ ph	r = 0.83
acidity	~ residual_sugar	r = 0.69
acidity	~ ph	r = -0.74
price_eur	~ rating	r = 0.85

.. 10. PER-ROW FLAG SUMMARY

Clean rows : 16 / 20
Flagged rows: 4 / 20

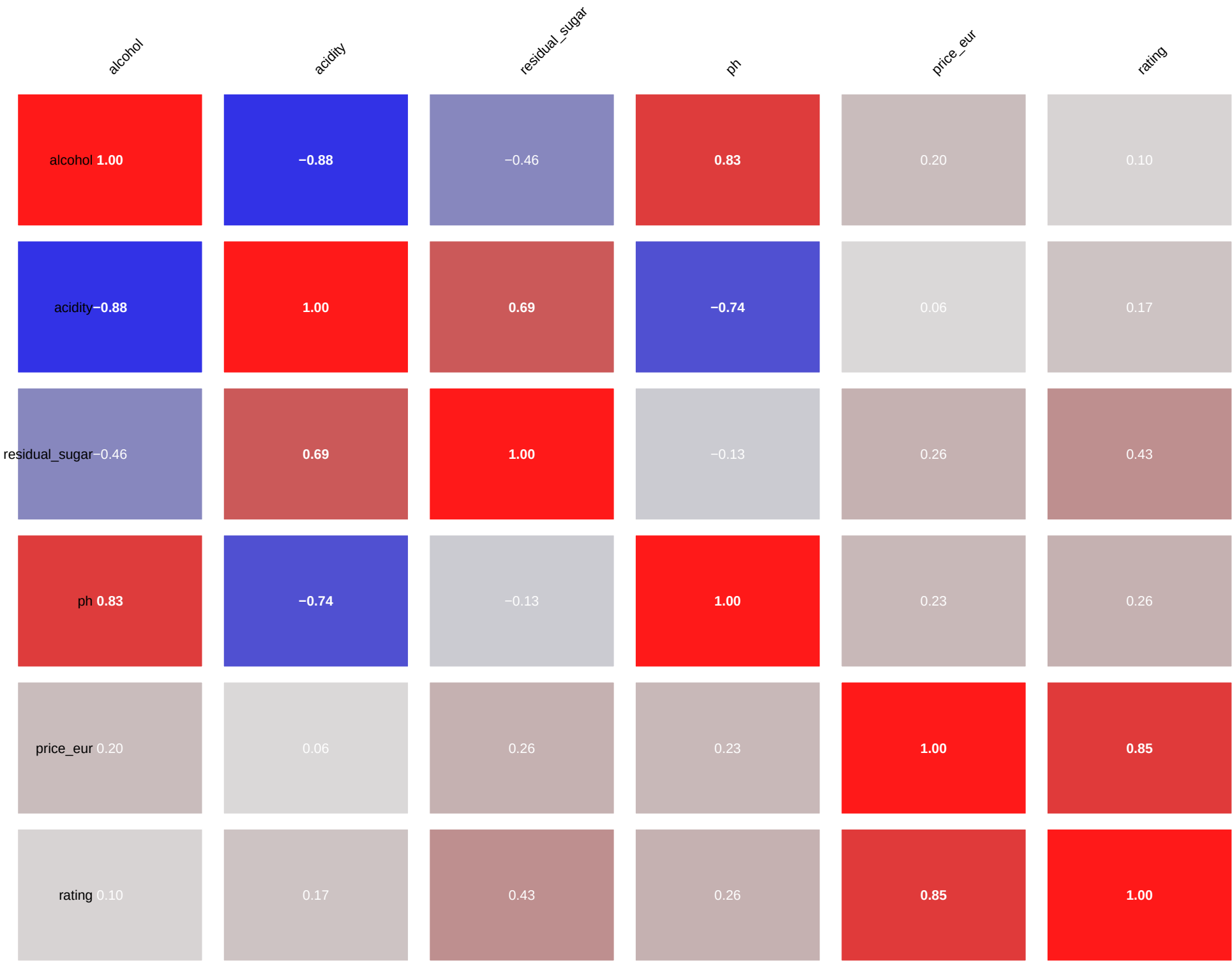
Flagged wines:

Row 3	Mosel Kabinett	[alcohol_outlier; residual_sugar_outlier]
Row 9	Napa Reserve	[price_eur_outlier]
Row 15	Tokaji Aszu	[acidity_outlier; residual_sugar_outlier]
Row 19	Alsace Gewurz	[residual_sugar_outlier]

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QC REPORT COMPLETE
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Correlation Matrix



Red = positive | Blue = negative | Bold = |r| > 0.6